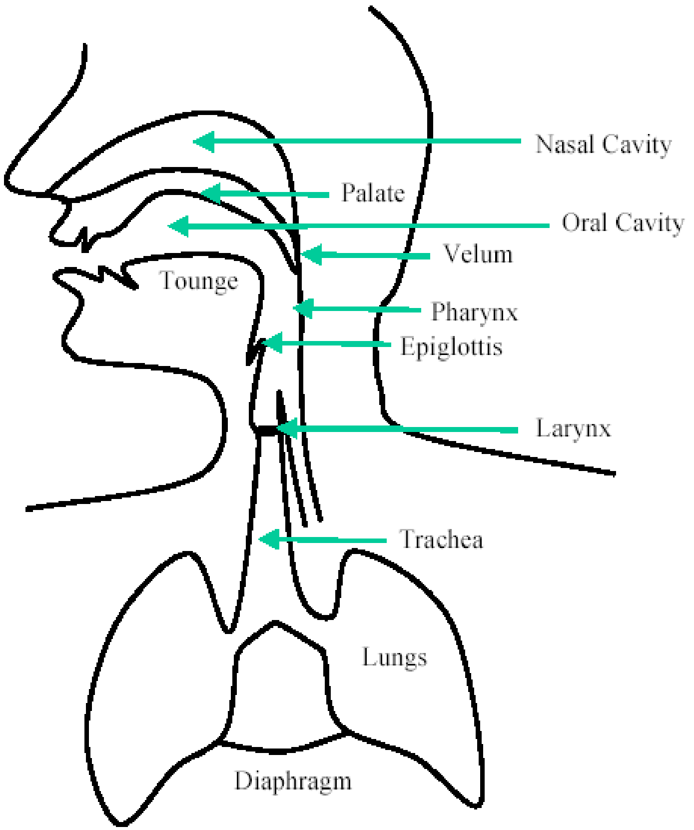
Lec 1:
Speech Generation and Perception

Speech Generation and Perception

- The study of the anatomy of the organs of speech is required as a background for articulatory and acoustic phonetics.

- An understanding of hearing and perception is needed in the field of both speech synthesis and speech enhancement and is useful in the field of automatic speech recognition.

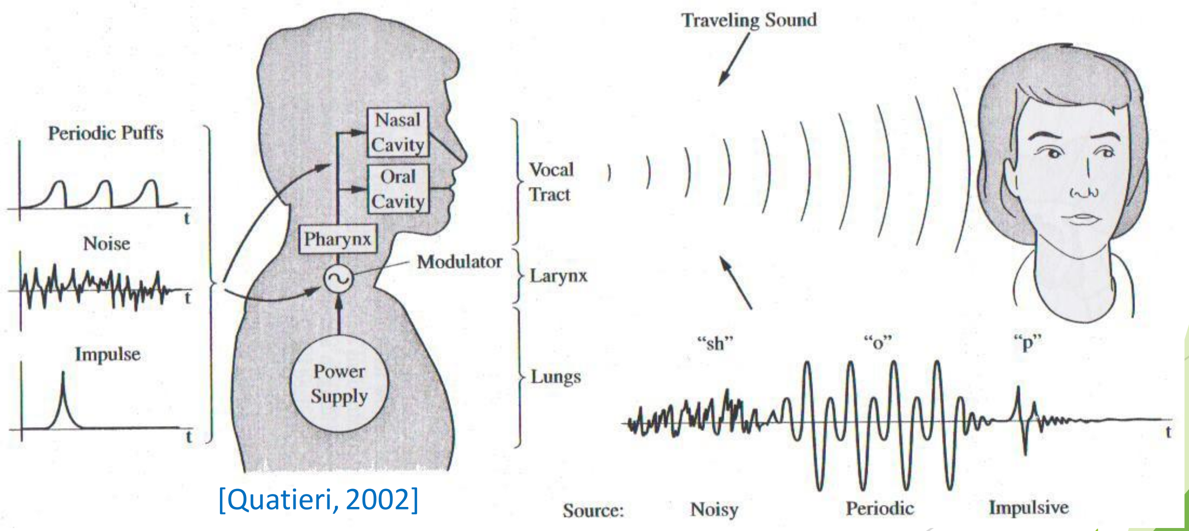
Schematic diagram of the human speech production



Anatomy of the speech organs

The speech organs can be broadly divided into three groups

- Lungs: serve as a “power supply” and provides airflow to the larynx
- Vocal chords (Larynx): modulate the airflow into either a periodic sequence of puffs or a noisy airflow source
  - A third type of source is impulsive
  - Exercise, say the word “shop” and determine where each sound occurs
- Vocal tract: converts modulated airflow into spectrally “colored” signal



[Quatieri, 2002]

Organs of Speech

- Lungs and trachea(ششها و نای)
  - source of air during speech.
  - The vocal organs work by using compressed air; this is supplied by the lungs and delivered to the system by way of the trachea.
  - These organs also control the loudness of the resulting speech.
  - The trachea and lungs together constitute the *pulmonary tract*.

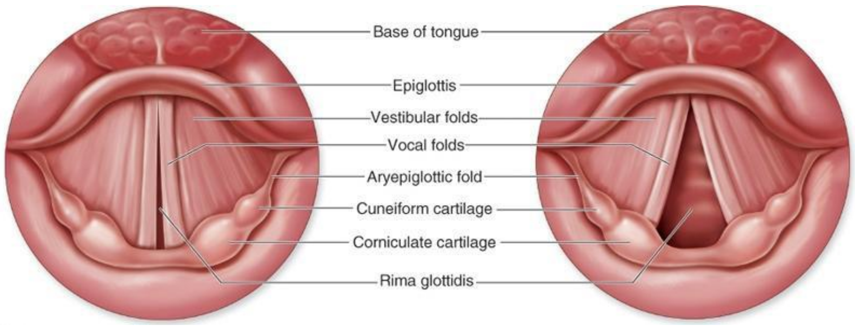
Organs of Speech

- The Larynx (حنجره)
  - This is a complicated system of cartilages and muscle containing and controlling the vocal cords. Principle parts are :
    - Cricoid cartilage
    - Thyroid cartilage
    - Arytenoid cartilage
    - Vocal cords
  - The place where the vocal folds come together is called the *glottis*.

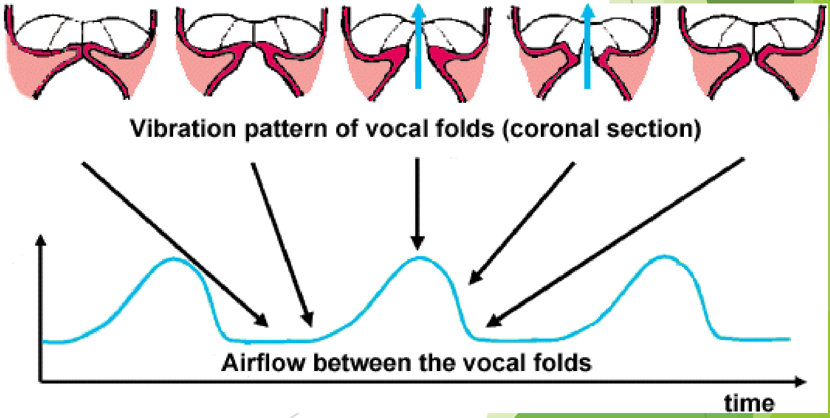
Organs of Speech: The vocal folds

Two masses of flesh, ligament and muscle across the larynx

- Fixed at the front of the larynx but free to move at the back and sides
- Can be in one of three primary states
  - Breathing: Glottis is wide, muscles are relaxed, and air flows with minimal obstruction
  - Voicing: vocal folds are tense and are brought up together. Pressure builds up behind, leading to an oscillatory opening of the folds
    Unvoiced: similar to breathing state, but folds are closer, which leads to turbulences (i.e. aspiration, as in the sound *h* in 'he') or whispering



http://academic.kellogg.edu/herbrandsonc/bio201_mckinley/f25-5b_vocal_folds_lary_c.jpg

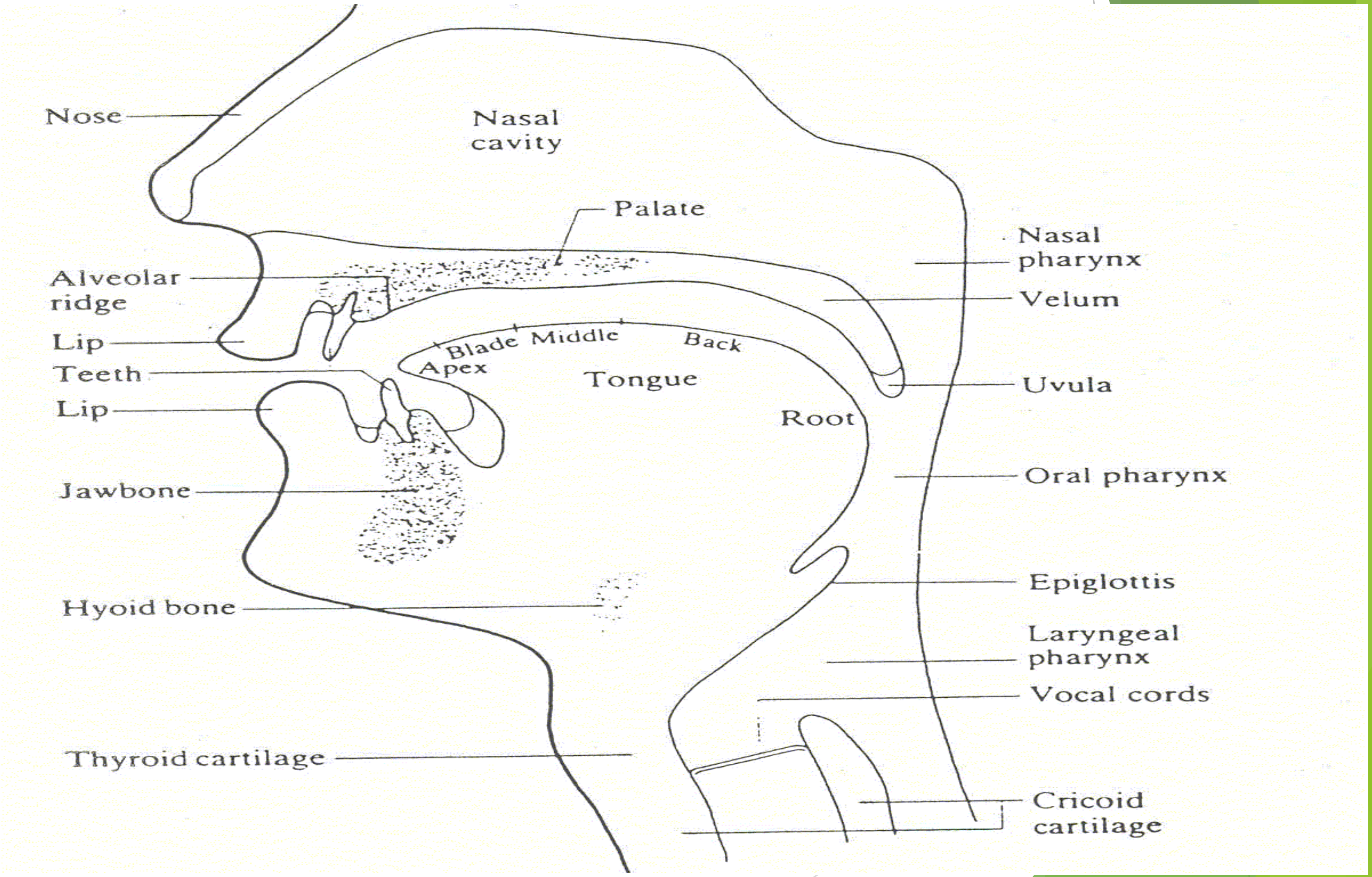


http://biorobotics.harvard.edu/research/heather2.gif

Organs of Speech

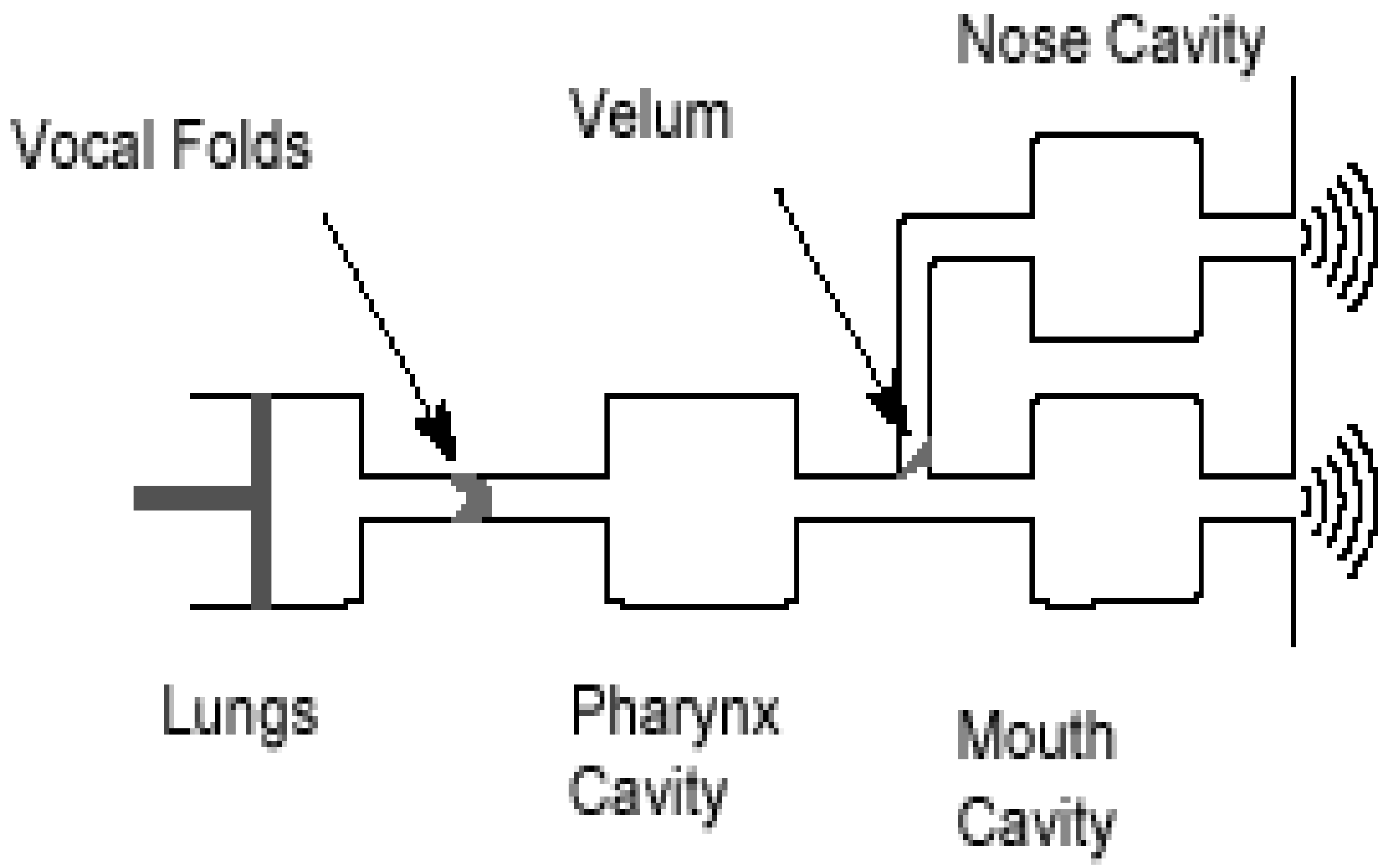
- The Vocal Tract (جهاز صوتی)
  - Laryngeal pharynx
    - beneath epiglottis
  - Oral pharynx
    - behind tongue, between epiglottis and velum (نرم کام)
  - Nasal pharynx
    - Above velum, rear end of nasal cavity
  - Oral cavity (حفره دهانی)
    - Forward of the velum and bounded by lips, tongue and palate
  - Nasal cavity (حفره بینی)
    - Above the palate and extending from the pharynx to the nostrils

Vocal Tract



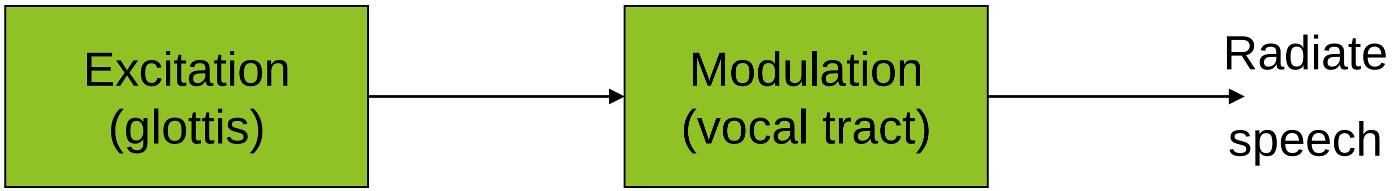
Nose
Nasal cavity
Palate
Alveolar ridge
Lip
Teeth
Lip
Jawbone
Hyoid bone
Thyroid cartilage
Nasal pharynx
Velum
Uvula
Blade
Middle
Back
Apex
Tongue
Root
Oral pharynx
Epiglottis
Laryngeal pharynx
Vocal cords
Cricoid cartilage

Vocal Tract Model



Speech Production

- The operation of the system is divided into two functions :
  - Excitation
  - Modulation



Speech Production

- Excitation :is done in several ways
  - Phonation (making of a voiced sound)
    - This is the oscillation of the vocal cords
    - The arytenoid cartilages close and stretch the vocal cords
    - When air forced through the vocal, they vibrate
    - The opening and closing of the cords breaks the airstream up into pulses

Speech Production

- The repetition rate of the pulses is termed ***pitch.***

- At low levels of air pressure oscillation may become irregular, this irregularities are known as “vocal fry”.

- Speech sounds accompanied by phonation are called voiced (واکدار); others, unvoiced (بیواک) or mute.

- Whispering (speak softly)
  - The vocal cord are drown together, but with small triangular opening between arytenoid cartilages

Speech Production

- Frication (سايش)
  - Frication can occur with or without phonation

- Compression
  - If the release is abrupt and clean, the sound is a ***stop*** or ***plosive***

  - If gradual and turbulent, the sound can pass into the related fricative and is termed an affricative

Speech Production

- Vibration
  - If air is forced through a closure other than the vocal cords, vibrations may be set up
- Modulation
  - This is what we do to impose information on the glottal output
    - Articulatory phonetics: how the organs of speech are positioned to produce any given speech sound
    - Acoustic phonetics: what the measurable acoustical correlates of any given speech sound are and how acoustical features in general correspond to phonetic and articulatory ones